Deep Literature Review and Architectural Analysis of SpectraLoRA: Equivariant Adaptive Alignment of Molecular Foundation Models

The intersection of quantum chemistry, equivariant deep learning, and high-throughput distributed systems has catalyzed a paradigm shift in atomistic modeling. Specifically, the development of compact, parameter-efficient graph foundation models capable of predicting complex multidimensional properties—such as Infrared (IR) and Raman spectra—represents the frontier of computational materials science. The following exhaustive literature review contextualizes the architectural, physical, and infrastructural claims surrounding the development of a ~7M parameter equivariant graph neural network designed for molecular spectroscopy. This architecture uniquely integrates Equivariant Low-Rank Adaptation (ELoRA), adaptive rank allocation (AdaLoRA), and rank decomposition to act as a highly specialized foundation model. Furthermore, the model is designed to undergo post-hoc alignment to bridge the gap between theoretical Density Functional Theory (DFT) calculations and empirical experimental conditions, utilizing a distributed training pipeline orchestrated via Ray and Amazon FSx for Lustre.

Following a rigorous analysis of the quantum mechanical foundations, the polarizability bottleneck in spectroscopy, the mathematics of equivariant parameter-efficient fine-tuning (PEFT), and distributed data engineering, this report will culminate in a critical evaluation of potential manuscript titles for submission to the Conference on Neural Information Processing Systems (NeurIPS), directly addressing the optimal framing for such an architecture.

First Principles: The Quantum Mechanical Foundations of Molecular Modeling

To fully appreciate the architectural requirements of a neural network tasked with mapping molecular graphs to quantum properties and vibrational spectra, one must first establish the physical laws the model acts as a surrogate for. The fundamental behavior of any atomistic system is governed by the non-relativistic Schrödinger equation, which provides the complete quantum-mechanical description of interacting particles.[1] Within this framework, observables—dynamical variables that can be physically measured—are represented by linear Hermitian operators that satisfy the principle of superposition.[1]

For a molecular system, the total energy is defined by the Hamiltonian operator (H), which is the sum of the kinetic energy (T) and potential energy (V) operators.[1] Formally, the time-independent molecular Hamiltonian ($\hat{H}_{mol}$) is expressed as the sum of the kinetic energy of the nuclei, the kinetic energy of the electrons, and the various electrostatic interactions between them (electron-nucleus, nuclear-nuclear, and electron-electron).[1] Because the exact determination of the total molecular wavefunction (Ψ) depends simultaneously on the spatial and spin coordinates of every particle in the system, solving the Schrödinger equation directly is computationally intractable for all but the simplest molecules.[1]

To render the molecular problem solvable, computational chemistry universally relies on the Born-Oppenheimer (BO) approximation. The BO Ansatz posits that due to the massive disparity in mass between heavy nuclei and light electrons, the kinematics of the two can be effectively decoupled.[1] The electrons are assumed to adjust instantaneously to any change in the nuclear configuration. Consequently, the non-separable Hamiltonian is factorized into two subsystems: a fast-moving electronic subsystem ($\hat{H}_{fast}$) and a slower-moving nuclear subsystem ($\hat{H}_{slow}$).[1] This factorization yields the classic expression for the total wavefunction: $\Psi(r, R, t) = \sum_l \Phi_l(r; R)\chi_l(R, t)$ where $\chi_l(R, t)$ represents the nuclear wavefunctions and $\Phi_l(r; R)$ represents the electronic wavefunctions parameterized by a fixed nuclear configuration R.[1]

Despite the BO approximation, explicitly solving for the complex $3N$-dimensional many-body electronic wavefunction encounters an exponential wall when the number of atoms exceeds a critical threshold.[1] Density Functional Theory (DFT) circumvents this dimensionality catastrophe by relying on the Hohenberg-Kohn theorems.[1] The first Hohenberg-Kohn theorem proves that the exact ground-state energy and all related properties of a many-electron system are uniquely determined by its 3D spatial electron density, $n(r)$, rather than the highly complex multidimensional wavefunction.[1] The second theorem establishes that a universal functional for the energy $E[n]$ can be defined, and the density that minimizes this functional is the exact ground-state density.[1]

The practical implementation of DFT utilizes the Kohn-Sham formalism, which maps the interacting many-body electron problem onto a fictitious auxiliary system of non-interacting electrons that are constructed to reproduce the exact ground-state density of the true system.[1] The quantum mechanical properties of the original system, including many-body effects, are captured in the Kohn-Sham system via a multiplicative potential known as the exchange-correlation functional ($E_{xc}[n]$).[1]

Once the accurate electron density is determined via DFT, computational chemists can access precise bonded and non-bonded interatomic interactions by way of the Hellmann-Feynman theorem.[1] This theorem proves that the derivative of the total energy with respect to a parameter P (such as the spatial coordinates of a nucleus) is exactly equal to the expectation value of the derivative of the Hamiltonian with respect to that same parameter: $\frac{dE}{dP} = \langle \frac{\partial H}{\partial P} \rangle$ While DFT provides a mathematically rigorous framework for calculating the potential energy surface (PES) and deriving interatomic forces, the required Self-Consistent Field (SCF) iterations scale poorly—typically $O(N^3)$ or $O(N^4)$ with respect to system size.[1] Consequently, evaluating high-order derivatives, such as the mass-weighted Quantum Hessian, or running high-throughput molecular dynamics over vast conformational spaces remains a severe computational bottleneck.[1] Machine learning interatomic potentials (MLIPs) and graph foundation models are specifically designed to bypass these iterative DFT calculations by learning the direct non-linear mapping from atomic coordinates to the PES and related tensorial properties.[1]

Equivariant Graph Neural Networks as Quantum Surrogates

When replacing quantum mechanical operators with deep neural networks, the architecture must strictly preserve the physical symmetries inherent to 3D space. If a molecule is rotated or translated, its scalar properties (such as total energy) must remain invariant, while its vectorial properties (such as dipole moments and atomic forces) and tensorial properties (such as polarizability and the Hessian matrix) must rotate codirectionally with the input coordinates.[1] Standard message-passing neural networks (MPNNs) that operate solely on invariant pairwise distances fail to capture the high-order angular and directional information necessary to accurately model complex molecular fields.[5]

Equivariant Graph Neural Networks (EGNNs) have subsequently emerged as the standard for atomistic modeling.[3] These architectures strictly enforce geometric consistency by constraining neural operations to the Special Orthogonal group SO(3) or the Euclidean group E(3).[5] They achieve this by representing node features as coefficients of spherical harmonics and utilizing Clebsch-Gordan tensor products to manage the interactions between features of different angular momenta during message passing.[5]

Several specialized architectures have been developed to manage the high computational overhead of tracking high-degree geometric tensors, which provides essential context for the deployment of a highly compact 7M parameter foundation model.

Architecture	Equivariant Mechanism	Tensorial Handling & Primary Output	Strategy for Computational Efficiency
NequIP [3]	E(3)-equivariant convolutions	Predicts interatomic potentials and forces using geometric tensors up to a specified angular momentum l.	Highly data-efficient, but relies on massive message passing across deep layers, leading to high computational costs for inference.[3]
MACE [8]	Atomic Cluster Expansion (ACE)	Utilizes equivariant internal features to compute energy, forces, and Hessians via many-body messages.[8]	Computes high-order features via tensor products of lower-order features rather than explicit enumeration, drastically reducing required message-passing layers.[8]
DetaNet [10]	Deep Equivariant Tensor Attention	Broad spectrum predictions across scalars, vectors, and 2nd/3rd-order tensors (dipole, polarizability, Hessian).[10]	Splits massive target matrices into manageable subsets; treats atomic tensors and interatomic tensors with separate, focused attention models.[10]
Mo3ENet [12]	EGNN Encoder	Embeds input point clouds as fixed-dimensional equivariant representations for vector/tensor	Relies on fixed representations to reduce the dynamic expansion of tensor products, though occasionally

		property prediction.[12]	outperformed by purpose-built models like DetaNet.[12]

A 7M parameter graph model represents a highly compact architecture compared to the massively parameterized large language models (LLMs) typical of modern AI, but within the domain of equivariant molecular physics, it possesses sufficient capacity to act as a robust foundation model. The key to achieving high expressivity with a small parameter count lies in the network's ability to efficiently handle the tensor product expansions.[4] Models like DetaNet are explicitly tailored for the simulation of organic molecular spectra, proving that a targeted EGNN can match quantum-chemical accuracy on benchmark datasets like QM9S.[10]

A critical architectural challenge for these models is the prediction of the Quantum Hessian. The Hessian is a $(3N)^2$ matrix of second derivatives of the potential energy with respect to nuclear positions, dictating the local curvature of the PES.[1] Directly predicting this massive matrix generally overwhelms compact networks. DetaNet solves this by splitting the prediction task: it uses separate sub-models to compute the isolated atomic tensors and the pairwise interatomic tensors, analytically recombining them to form the full dynamical matrix.[10] This structural methodology enables a compact 7M parameter model equipped with multiple task heads to predict high-order tensors effectively without experiencing capacity exhaustion.[10]

The Physics of Vibrational Spectroscopy: Hessian Success and the Polarizability Bottleneck

The explicit goal of the proposed 7M parameter foundation model is to map molecular graphs directly to IR and Raman spectra. The theoretical feasibility of this relies on the physical mechanisms driving vibrational spectroscopy, which exhibit distinct behavior when modeled by EGNNs.

Vibrational frequencies—the positions of the peaks on a spectrum—are derived directly from the eigenvalues of the mass-weighted Quantum Hessian.[1] Because the Hessian is defined by the local harmonic restoring forces between atoms, graph neural networks (which fundamentally aggregate local neighbor information via message passing) are exceptionally adept at learning it.[1] Zero-shot evaluations of foundation models structured similarly to SpectraLoRA demonstrate unprecedented generalization of the Quantum Hessian across out-of-distribution (OOD) chemical phase spaces.[1] In blind evaluations against DFT benchmarks, neural surrogates accurately placed 60.6% of all predicted modes within a stringent $\pm 10\ cm^{-1}$ margin, achieving a median matched-line error of just $3.56\ cm^{-1}$.[1] Statistical validation confirmed a complete absence of systematic positional bias, meaning the MLIP preserved the isotropic harmonic restoring forces globally without artificially red-shifting or blue-shifting the frequencies.[1] This sub-$4\ cm^{-1}$ accuracy establishes compact EGNNs as highly reliable replacements for standard DFT in thermochemical pipelines, allowing for the direct extraction of Zero-Point Vibrational Energy (ZPE) and vibrational entropy.[1]

However, accurately predicting the positions of the spectral peaks is only half the challenge; a simulated spectrum must also accurately predict peak intensities (amplitudes). This is where compact foundation models encounter a severe physical limitation known as the "polarizability bottleneck".[1]

In Raman spectroscopy, the inelastic scattering of light is governed by the dynamic electronic response of the molecule to an external electromagnetic field.[1] The intensity of a Raman peak corresponding to a specific normal mode q_i is directly proportional to the derivative of the macroscopic polarizability tensor with respect to that nuclear displacement: $\left(\frac{\partial \alpha}{\partial q_i}\right)$.[1] Conversely, IR intensities depend on the derivative of the dipole moment.[17]

Unlike the local restoring forces that govern the Hessian, polarizability is an intrinsically non-local property. It reflects the global delocalization of the many-body wavefunction and the instantaneous fluctuation of the entire electron density across the macromolecule.[1] EGNNs, which construct representations through localized message passing, inherently struggle to capture these macro-scale wave-function delocalizations.[1] As documented in foundation model evaluations, while the Hessian frequencies achieved a median error of $3.56\ cm^{-1}$, the predicted Raman intensities (polarizability derivatives) exhibited a massive variance, yielding a Mean Absolute Error (MAE) of 0.85 in $\log_{10}$ units against DFT benchmarks.[1] An error of 0.85 on a logarithmic scale translates to a multiplicative physical variance by a factor of roughly seven, effectively destroying the predictive reliability of spectral amplitudes.[1]

This polarizability bottleneck causes intensity-weighted peak coverage to drop precipitously in the critical 400 to 1600 cm^{-1} fingerprint region, rendering raw model outputs invalid for direct experimental Raman fingerprint matching.[1] To bypass this limitation, contemporary architectures train auxiliary task heads explicitly focused on learning the first derivatives of polarizability and dipole moments alongside the Hessian.[6] However, because non-local electron density fluctuations are difficult to learn from static DFT datasets, the model must undergo rigorous post-training alignment to empirical experimental conditions.

Equivariant Parameter-Efficient Fine-Tuning (PEFT) in SO(3) Space

The user's architecture utilizes a highly specific fine-tuning stack—combining Equivariant LoRA

(ELoRA), Adaptive LoRA (AdaLoRA), and rank decomposition—to align the 7M parameter base model to experimental IR/Raman targets. Understanding why this specific combination is necessary requires an analysis of the mathematical constraints of SO(3) parameter spaces.

As pre-trained interatomic potentials scale, full-parameter fine-tuning becomes computationally expensive and frequently induces catastrophic forgetting, wherein the model loses its generalized pre-trained understanding of fundamental chemical interactions when overfitted to a specific downstream task (such as predicting a specific experimental Raman dataset).[4] Parameter-Efficient Fine-Tuning (PEFT) addresses this by freezing the majority of the pre-trained weights and injecting a small number of trainable parameters.[4]

The most common PEFT methodology is Low-Rank Adaptation (LoRA), which decomposes weight updates into two low-rank matrices ($\Delta W = A \times B$).[7] However, applying standard LoRA directly to an SO(3) Equivariant Message Passing Neural Network breaks the network's equivariance.[20] Equivariance in these models is maintained by strictly controlling how geometric tensors (represented by spherical harmonics) interact.[7] A global low-rank projection indiscriminately mixes features across different angular momentum degrees, destroying the carefully constructed Clebsch-Gordan tensor products and voiding the physical symmetry guarantees.[20]

Path-Dependent Decomposition: The ELoRA Mechanism

To preserve equivariance during adaptation, the Equivariant Low-Rank Adaptation (ELoRA) framework introduces a path-dependent decomposition strategy.[4] The equivariance of a fully connected tensor product stems from its constraints on the specific mathematical paths between input and output tensors.[4] ELoRA restricts the low-rank updates by decomposing the weights strictly along each valid interacting path rather than applying a global decomposition.[4]

By projecting equivariant messages into a lower-dimensional space only along symmetry-preserving routes, ELoRA ensures that the updated representations remain physically valid.[7] Empirical benchmarks demonstrate the superiority of this approach: on the rMD17 organic dataset, ELoRA achieved a 25.5% improvement in energy prediction accuracy and a 23.7% improvement in force prediction accuracy compared to full-parameter fine-tuning, while maintaining the exact inference latency of the base model.[7]

Adaptive Parameter Allocation: AdaLoRA and Rank Decomposition

In a multi-head foundation model, where a central graph encoder feeds into several diverse prediction heads (e.g., predicting the Hessian, the polarizability, the dipole moment, and other quantum properties), applying a uniform rank across all layers and heads is highly inefficient.[22] The user explicitly noted that their 7M parameter model possesses "other task heads that I don't bother messing with much." Standard LoRA would blindly allocate trainable parameters to these inert task heads, wasting the limited parameter budget.

Adaptive Low-Rank Adaptation (AdaLoRA) solves this by dynamically adjusting the rank of the update matrices (ΔW) based on real-time importance scoring during the fine-tuning process.[23] AdaLoRA utilizes Singular Value Decomposition (SVD) to parametrically fine-tune the singular values of the weight matrices.[24] It calculates a sensitivity score—often utilizing Fisher information, which measures the expected sensitivity of the model output to parameter perturbations via the gradients of the log-likelihood—and applies exponential smoothing to stabilize the signals.[25]

By incorporating AdaLoRA into the ELoRA framework, the network actively prunes the rank of the adapters associated with the ignored task heads while expanding the rank capacity of the adapters connected to the active IR/Raman projection layers.[25] Furthermore, analyses of layer-wise perturbation demonstrate that earlier layers in an EGNN typically act as system-dependent spatial feature extractors that must adapt to new data distributions, whereas deeper layers encode common, highly stable transformations.[26] AdaLoRA autonomously recognizes this heterogeneous sensitivity, allocating heavier low-rank matrices to the early encoding layers and pruning the deeper layers.[26]

This synthesis of ELoRA (for symmetry preservation) and AdaLoRA (for surgical parameter budgeting) allows a compact 7M parameter model to behave with the adaptability of a much larger network. It establishes a highly specialized, task-aware fine-tuning pipeline that optimally redirects gradient flows toward the specific goal of spectroscopy prediction without destabilizing the foundational physics learned during pre-training.

Post-Training Alignment: Bridging the Sim-to-Real Gap in Spectroscopy

The user's methodology requires performing "post-hoc align[ment] to experimental conditions." This phase is critical because foundation models pre-trained on massive synthetic datasets (such as QM9S or artificially generated DFT trajectories) inevitably inherit the biases and approximations of the underlying quantum theory.[1]

Standard DFT calculations, particularly those utilized to generate massive pre-training corpora, typically evaluate the PES within the harmonic approximation.[16] The harmonic approximation assumes that interatomic forces behave as perfect springs. While this is sufficient for identifying the topological ground state, it systematically fails to capture anharmonic vibrational effects, which play a crucial role in physical systems at finite temperatures or in complex solvent environments.[16] Consequently, theoretical Raman and IR spectra generated from harmonic Hessians consistently exhibit positional shifts and amplitude errors when compared to empirical, real-world spectroscopic measurements.[27]

This discrepancy is known as the "sim-to-real" gap.[1] To bridge this divide without recalculating the entire dataset using prohibitively expensive Vibrational Self-Consistent Field (VSCF) corrections, the model undergoes post-training alignment.[27] During this phase, the behavior of the pre-trained EGNN is explicitly biased toward experimental ground truths.[30]

Alignment in computational chemistry utilizes structured reward modeling. Models are fine-tuned against high-resolution experimental spectral databases (e.g., NIST, PNNL, or AIST).[31] Advanced alignment frameworks move beyond simple scalar loss functions by utilizing multi-dimensional, contextual rubrics.[32] For example, Alternating Reinforcement Learning with Rubric Rewards (ARL-RR) optimizes different physical criteria sequentially, preventing the variance contraction that occurs when disparate errors (such as a frequency shift vs. an amplitude suppression) are linearly compressed into a single scalar reward.[32] Similarly, Force-Energy Disentangled (FED) alignment ensures that corrections to the predicted potential energy do not inadvertently corrupt the mechanical equilibrium defined by the interatomic forces.[33]

By executing this experimental alignment via the ELoRA + AdaLoRA framework, the network essentially learns a parameter-efficient mapping function that corrects the theoretical harmonic biases. The low-rank adapters absorb the complex, anharmonic empirical noise (such as solvent interactions or temperature-dependent peak broadening), transforming the deterministic DFT surrogate into a physically accurate predictor of real-world experimental spectroscopy.[16]

Distributed Systems Architecture: Ray and Amazon FSx for Lustre

Training and aligning molecular foundation models on complex graph-structured data introduces severe infrastructural bottlenecks. The offline generation of ab initio quantum data, the featurization of molecular graphs, and the evaluation of massive $(3N)^2$ Hessians are intrinsically CPU-bound and heavily dependent on storage I/O.[1] Conversely, the training of the EGNN requires high-throughput, uninterrupted data streaming to maintain high utilization rates across distributed GPU clusters.[1] If the infrastructure fails to reconcile this impedance mismatch, the GPUs will stall while waiting for data, decimating training efficiency.[35]

The user's architecture circumvents this by deploying a distributed training pipeline orchestrated via Ray, utilizing Amazon FSx for Lustre as the shared storage bedrock.[1] This setup constitutes a sophisticated, decoupled four-plane architecture.[1]

The Storage Bedrock: FSx for Lustre

High-performance computing for cheminformatics demands parallel file systems capable of sustaining sub-millisecond latencies and immense bandwidth across thousands of concurrent compute nodes.[37] Amazon FSx for Lustre provides a fully managed, POSIX-compliant file system designed explicitly to eliminate storage bottlenecks in machine learning workloads.[1]

To fully saturate the computational capacity of modern accelerator instances (such as Amazon EC2 P4d or P5 instances), FSx for Lustre integrates with the Elastic Fabric Adapter (EFA) and NVIDIA GPUDirect Storage (GDS).[40] EFA bypasses the traditional operating system networking stack, utilizing the Scalable Reliable Datagram (SRD) protocol to optimize node-to-node data transfer.[40] Concurrently, GDS establishes a direct memory access path between the remote Lustre storage and the local GPU memory.[40] By bypassing the CPU bounce buffers entirely, this integration propels per-client throughput to an extraordinary 1,200 Gbps.[40] This ultra-high bandwidth ensures that the complex 3D graph structures and their associated quantum tensorial labels are ingested into GPU memory fast enough to prevent computational starvation.[35]

Data Engineering and Orchestration via Ray

Ray serves as the unified compute framework that orchestrates this infrastructure, effectively replacing manual MPI (Message Passing Interface) scripting with scalable Python-native APIs.[36] Ray natively handles both task-parallel workloads (such as offline data featurization) and actor-based computations (such as maintaining stateful distributed training workers).[43]

During the offline data engineering phase, raw molecular datasets are ingested and transformed into graph representations by a swarm of CPU workers.[1] To prevent locking contention on the shared FSx file system, the architecture utilizes a shared-nothing approach; workers generate local, ephemeral SQLite3 indices to map molecule identifiers to physical storage shards, guaranteeing high-throughput, lock-free materialization.[1]

For the training phase, Ray orchestrates the distributed PyTorch jobs utilizing Distributed Data Parallel (DDP) or Fully Sharded Data Parallel (FSDP) backends.[1] In an FSDP setup, the EGNN's model parameters, gradients, and optimizer states are sharded across the cluster's GPUs, effectively pooling the VRAM and allowing for the manipulation of high-dimensional tensor products without triggering out-of-memory errors.[45] Ray coordinates the critical NVIDIA Collective Communication Library (NCCL) primitives—such as AllGather and ReduceScatter—required to synchronize the shards during the forward and backward passes.[1]

To further optimize throughput over the FSx network mount, the training data loader streams pre-sharded, pre-randomized data using an IterableDataset mechanism.[1] By assigning deterministic, coarse-grained chunks of data to specific GPU ranks based on the world size, the system avoids generating millions of small, random read requests.[1] While this explicitly trades the mathematical purity of a perfect global per-epoch shuffle for sustained sequential I/O, the design ensures maximal GPU saturation and completely mitigates cross-rank coordination stalls.[1] Ray also provides fail-fast meta-scheduling, autonomously terminating divergent training trials and recycling GPU resources to maximize cost-efficiency.[1]

NeurIPS Title Formulation and Critical Evaluation

The central query driving this analysis concerns the optimal selection of a manuscript title for submission to the Conference on Neural Information Processing Systems (NeurIPS). The title must accurately encapsulate the highly specialized architecture discussed herein: a compact 7M parameter equivariant graph foundation model, enhanced by ELoRA and AdaLoRA, trained in a distributed Ray+FSx environment, and aligned post-hoc to experimental IR/Raman spectroscopy data.

The user proposed four initial candidates for evaluation:

Proposed Title	**Critique and Analysis**
1. ELoRA for Molecular Spectra: Equivariant Post-Training of Graph Foundation Models	**Analysis:** A strong, straightforward title. It highlights the primary methodology (ELoRA) and the application (Molecular Spectra). However, it omits the critical "adaptive" nature of the architecture (AdaLoRA/rank decomposition) and does not explicitly convey the "sim-to-real" experimental alignment phase, which is a major contribution of the work.
2. From Molecules to Spectra via Equivariant Post-Training	**Analysis:** This title is overly broad and lacks technical specificity. While "From X to Y" is a common trope, it fails to signal the sophisticated PEFT methodologies (ELoRA/AdaLoRA) or the foundation model aspect to the NeurIPS audience, who typically look for algorithmic innovation.
3. SpectraLoRA: Equivariant Low-Rank Post-Training for Molecular Graph Models	**Analysis:** Highly effective. The introduction of a concise, memorable acronym ("SpectraLoRA") aligns perfectly with NeurIPS naming

	conventions (e.g., QLoRA, AdaLoRA, GraphLoRA). It succinctly captures the domain and the algorithmic contribution. However, it slightly undersells the "adaptive rank decomposition" and the experimental alignment.
4. Task-Specialized Equivariant Post-Training for IR/Raman Molecular Spectroscopy	**Analysis:** Highly descriptive but overly verbose and somewhat clunky. The inclusion of "IR/Raman" is specific, but the title lacks the algorithmic punch of mentioning LoRA/PEFT mechanisms. It reads more like a chemistry journal title than a machine learning conference title.

Synthesis and Recommendation

When formulating a title for NeurIPS, the goal is to balance algorithmic innovation, architectural scale, and domain application. The manuscript's most significant contributions are the fusion of path-dependent equivariance (ELoRA) with dynamic parameter budgeting (AdaLoRA) to force a compact (7M param) multi-head foundation model to adapt to empirical experimental conditions (sim-to-real alignment).

Building upon the strong foundation of Option 3, the optimal title should integrate the concepts of "Adaptive" (to signal the AdaLoRA/rank decomposition mechanism) and "Alignment" (to signal the post-hoc experimental sim-to-real tuning).

Recommended Title:

"SpectraLoRA: Adaptive Equivariant Alignment of Compact Graph Foundation Models for Experimental Spectroscopy"

Justification:

1. **"SpectraLoRA"**: Retains the strong, memorable acronym that immediately signals the application of Low-Rank Adaptation to spectral data, anchoring the paper in the active PEFT discourse at NeurIPS.[1]
2. **"Adaptive Equivariant"**: Explicitly captures both the ELoRA (Equivariant) and AdaLoRA (Adaptive/Rank Decomposition) mechanisms.[7] It signals to reviewers that the method does not merely apply static low-rank matrices, but dynamically manages the parameter

budget while preserving SO(3)/E(3) physical symmetries.

3. **"Alignment"**: Upgrades the phrase "Post-Training" to "Alignment". In modern ML parlance, "Alignment" conveys the nuanced process of shifting a pre-trained model's latent space to match human preference or real-world experimental ground truths (the sim-to-real gap), which perfectly describes the post-hoc tuning to experimental IR/Raman conditions.[30]
4. **"Compact Graph Foundation Models"**: Acknowledges the 7M parameter scale. Calling it "compact" preempts reviewer criticisms regarding model size, framing the 7M parameter count as a deliberate architectural achievement in efficiency rather than a limitation.

Alternatively, if the authors wish to heavily emphasize the algorithmic fusion of ELoRA and AdaLoRA directly in the title, a secondary option would be:

"Equivariant Adaptive LoRA (EAdaLoRA) for Bridging the Sim-to-Real Gap in Molecular Spectroscopy"

Ultimately, the "SpectraLoRA" nomenclature provides the strongest branding. By clearly delineating the adaptive parameter-efficient mechanisms and the experimental alignment focus, the title accurately reflects the complex distributed training, quantum physics, and equivariant graph mathematics encapsulated within the architecture.

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